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Industry 4.0 Security

Securing IoT-Enabled Factories with Open-Source Tools and convergence of IoT, AI, robotics, and automation in modern industries.

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Exam Level: RQF Level 03

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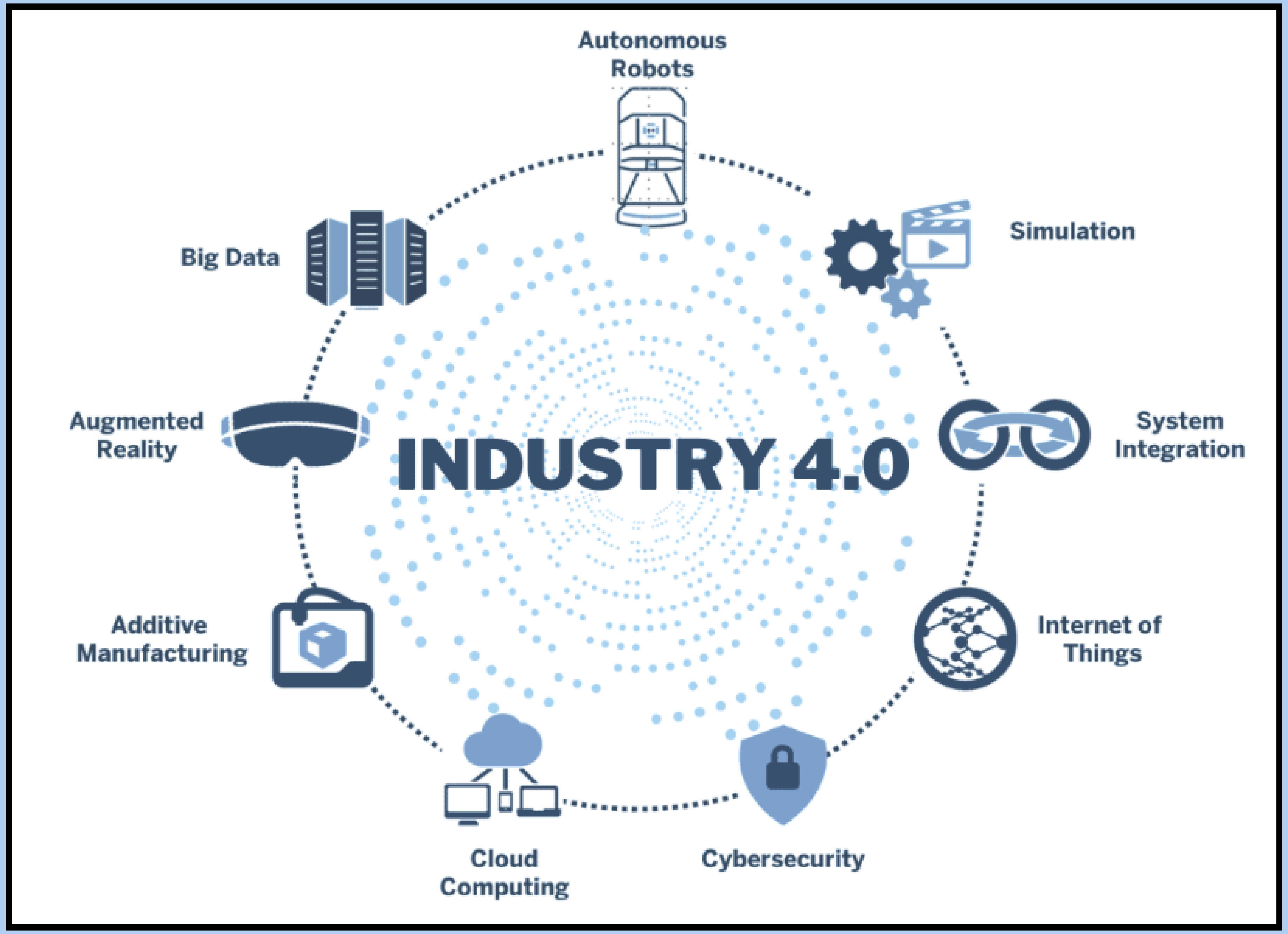
Introduction to Industry 4.0

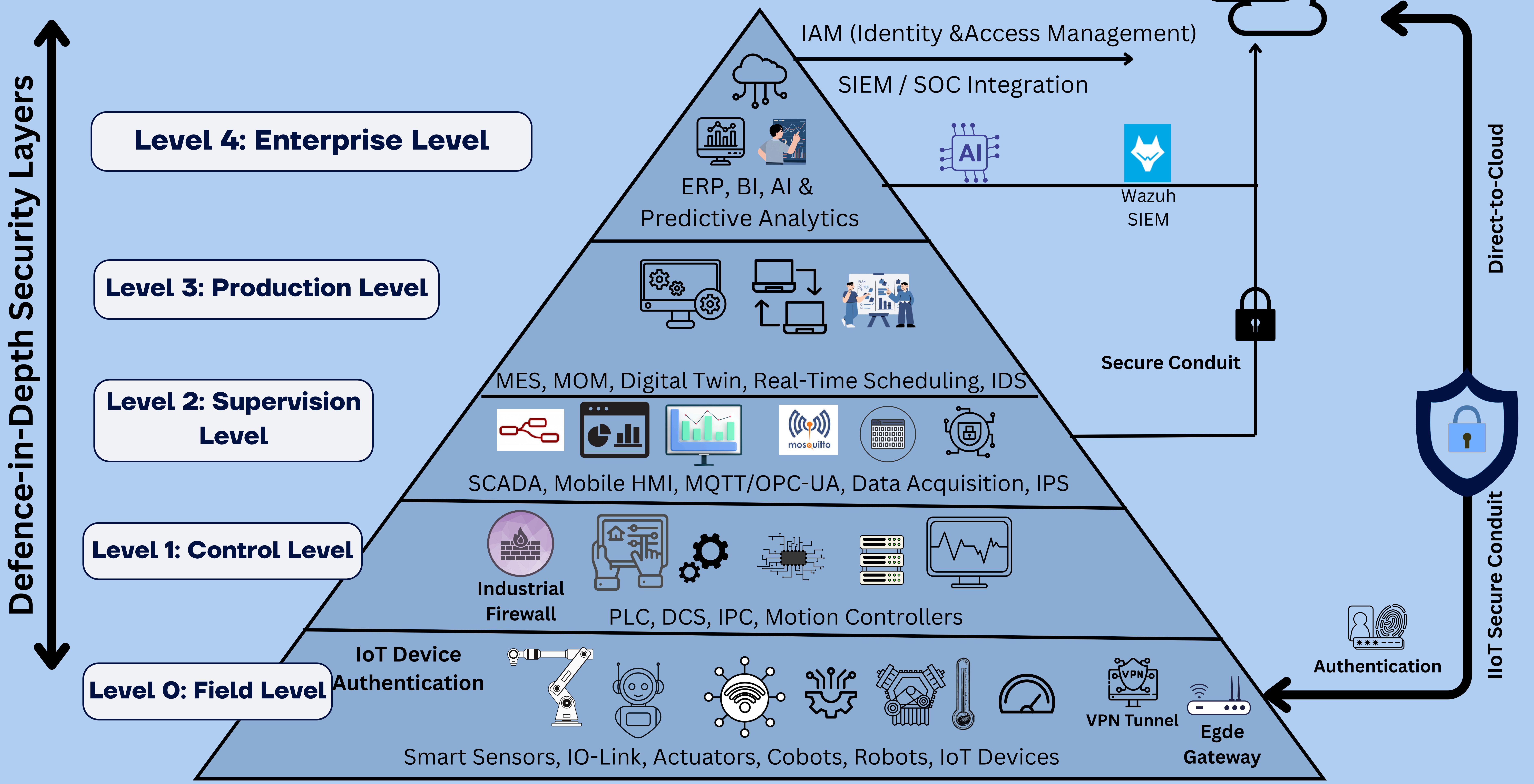
- **The Fourth Industrial Revolution (I4.0 or 4IR),** represents the intelligent networking of machines and processes in manufacturing using **IoT, AI, and Big Data.**

- ## IT-OT Integration:
- Merges **IT (**data, cloud, analytics**)** with **OT (**sensors, SCADA, machines**)**
- Enables real-time smart factory

- ## Key Aspects:
- Real-Time Visibility
- Resilience & Continuity
- Scalable Architecture
- Cyber-Physical Security
- Automation Intelligence



Industry 4.0 Secure Architecture



Cybersecurity Risks in IoT Factories

Real Threats Targeting Industry 4.0

Unauthorized Access

Attackers gain entry with invalid credentials

Malware

Malicious software targeting industrial control systems, disrupting factory operations and damaging OT equipment

Brute Force Attacks

Repeatedly attempts login credentials on connected devices and SSH ports

DDoS Attacks

Floods industrial servers with traffic, causing production systems and monitoring dashboards to go offline

Zero-day Exploits

Attackers target unknown vulnerabilities in IoT firmware and SCADA systems before patches are available

Man-in-the-Middle Attacks

Intercepts unencrypted communication between IoT sensors and control systems, allowing data manipulation

Source: ENISA Threat Landscape for Industrial Control Systems, 2022

Security Controls for Smart Factories

1.Data Encryption in Smart Factories

- **TLS** secures data in transit
- **AES-256** protects stored data
- **HSMs** manage encryption keys
- **Blockchain** shares keys securely

2.Authentication & Access Control

- **Zero Trust -** always verify
- Multi-Factor Authentication **(MFA)**
- **X.509** Digital Certificates
- Role-Based Access Control **(RBAC)**

3.Monitoring & Threat Detection

- **AI-Driven** anomaly detection
- **SIEM** aggregates OT/IT logs
- Network zone segmentation
- **IDS** monitors MQTT, Modbus, OPC-UA

Open-Source Tools

Mosquitto MQTT

- **Publish-subscribe** IoT protocol
- **Secure** device-to-device communication
- Enables **authentication**
- Trusted by Eclipse Foundation

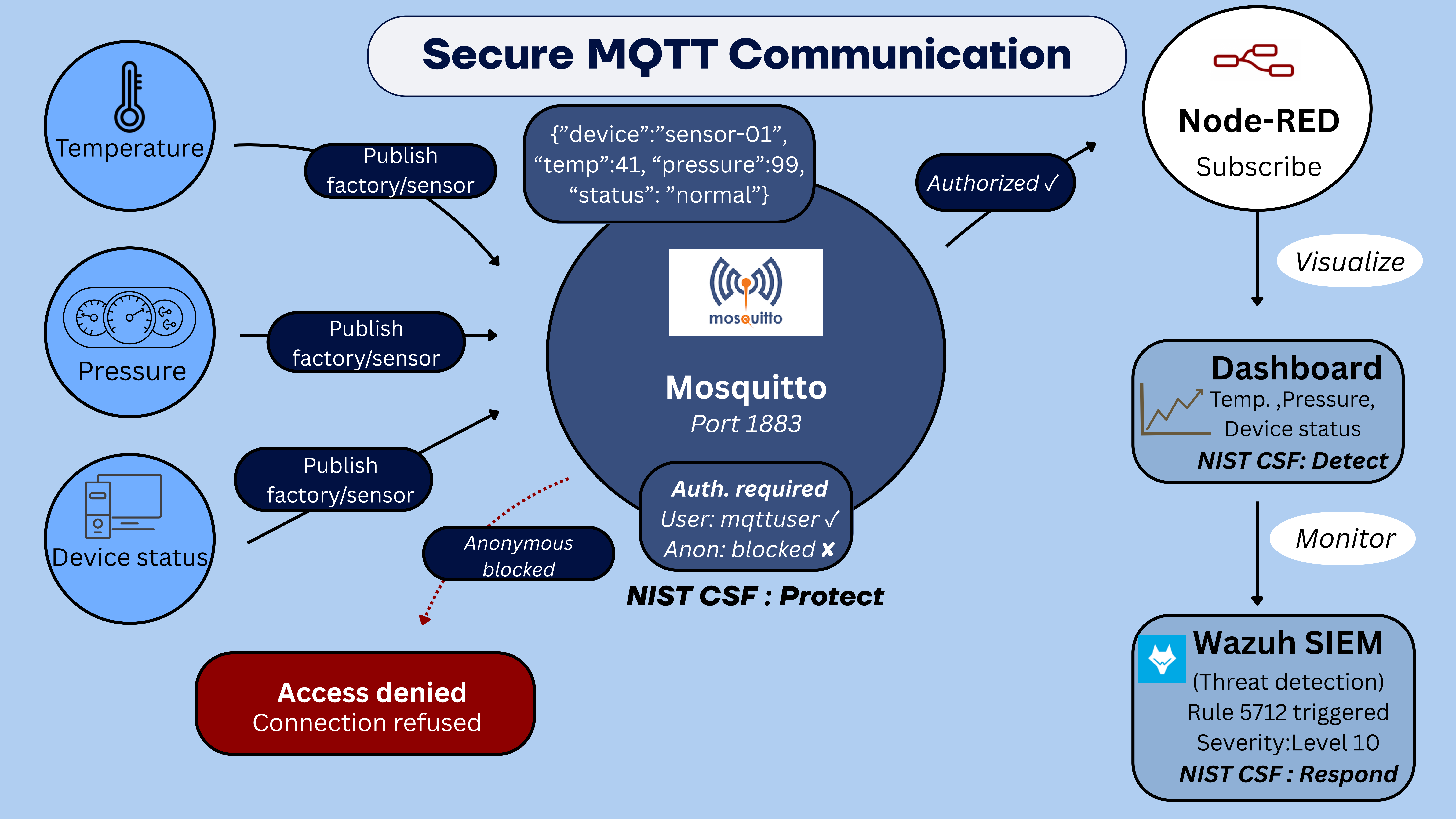
Node-RED

- **Flow-based** automation tool
- **Bridges** sensors and digital systems
- **Converts** raw data into visuals
- IBM-backed open-source project

Wazuh SIEM

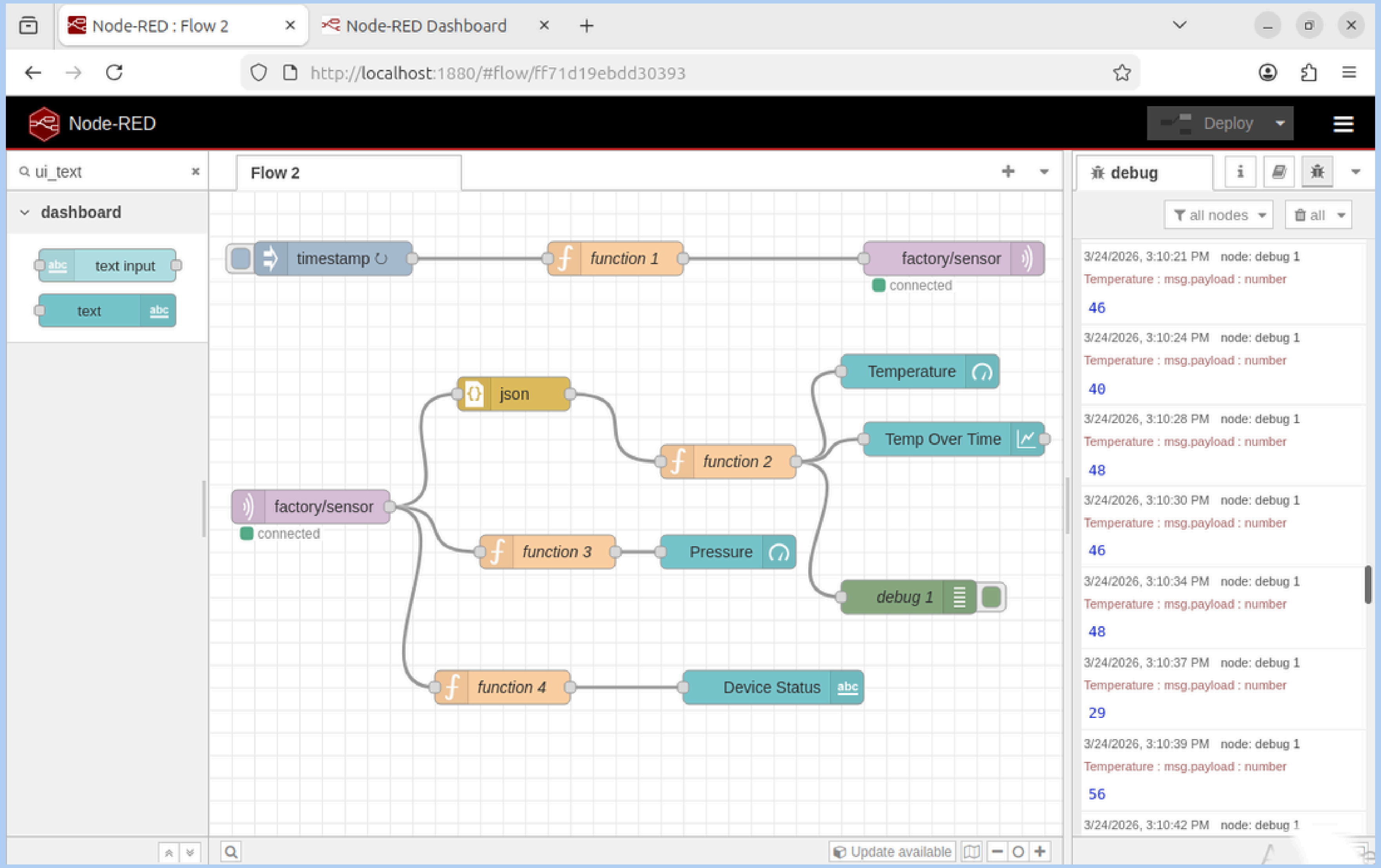
- Unified cyber threat **monitor**
- Detects suspicious activity
- Generates **severity-based** alerts
- **Flags threats** before damage occurs
- Globally recognized security platform

Source: mosquitto.org | nodered.org | documentation.wazuh.com

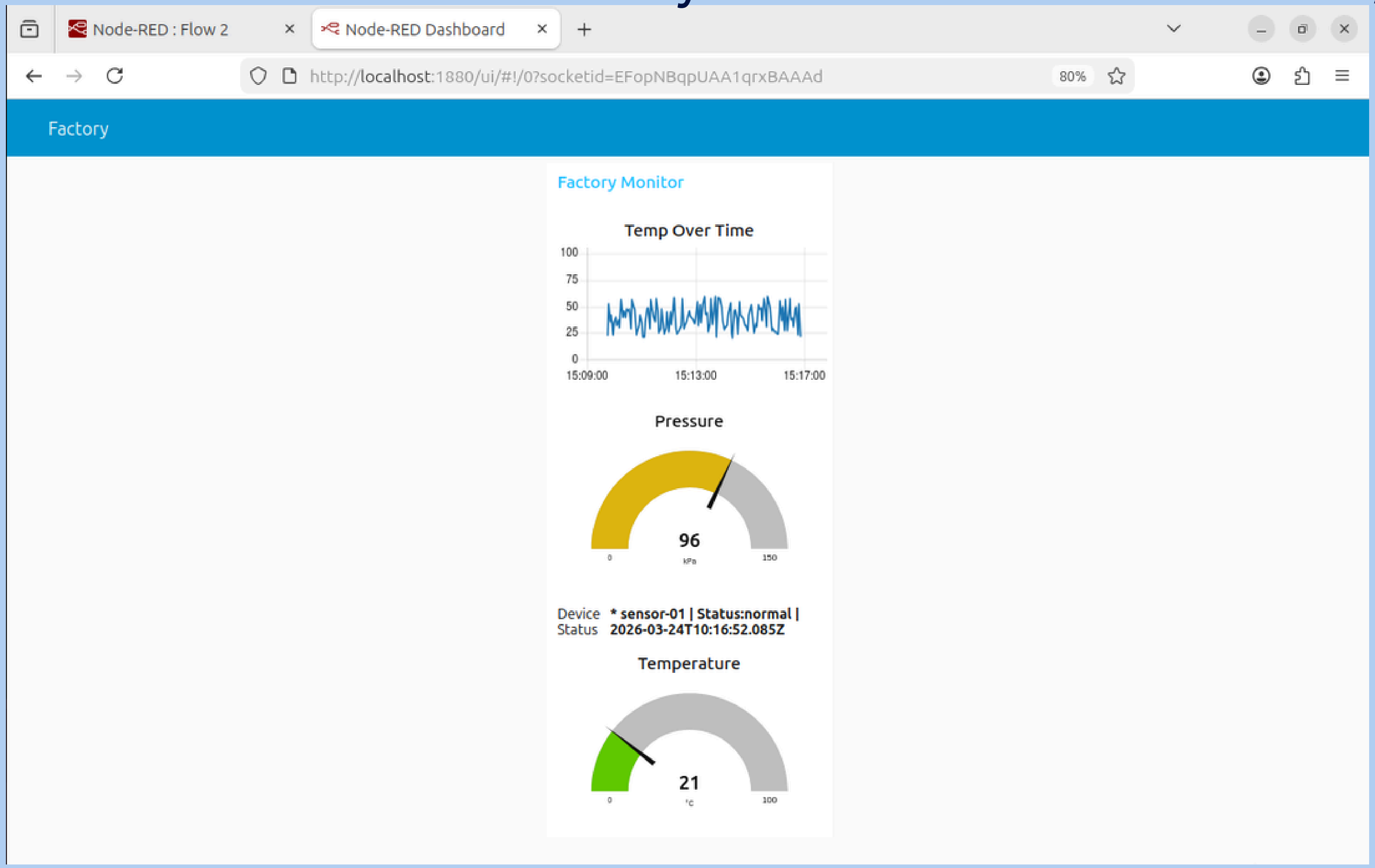

Secure MQTT Communication
Temperature
Pressure
Device status
Publish factory/sensor
Publish factory/sensor
Publish factory/sensor
{"device":"sensor-01", "temp":41, "pressure":99, "status": "normal"}
mosquitto
Mosquitto
Port 1883
Auth. required
User: mqttuser ✓
Anon: blocked ✘
NIST CSF : Protect
Anonymous blocked
Access denied
Connection refused
Authorized ✓
Node-RED
Subscribe
Visualize
Dashboard
Temp. ,Pressure, Device status
NIST CSF: Detect
Monitor
Wazuh SIEM
(Threat detection)
Rule 5712 triggered
Severity:Level 10
NIST CSF : Respond

Node-RED — Real-Time Industrial Monitoring

Node-RED Flow Canvas - localhost:



Factory Monitor Dashboard - localhost:1880/ui



What This Demo Proves

1 MQTT Secured

- Authenticated with username and password
- Port 8883 secured

2 Live Monitoring

- Temperature updated every 3 seconds
- Pressure monitored in real time
- Data from sensor-01

3 JSON Data Flow

- Structured sensor data sent over MQTT
- Topic: factory/sensor
- Parsed by Node-RED

4 NIST CSF: Detect

- Dashboard implements Detect function
- Anomalies visible instantly on screen
- Runs at localhost:1880/ui

IEC 62443 & NIST CSF Frameworks

Global Standards for Industrial Cybersecurity

IEC 62443

Industrial Security Standard

- Secure industrial automation and control systems
- Segments networks into zones and conduits
- Applies layered defence across all components
- Defines 4 security levels

NIST CSF

Risk-Based Framework

- **Identify -** Know your assets
- **Protect -** Safeguard systems
- **Detect -** Monitor continuously
- **Respond -** Act on threats
- **Recover -** Restore operations

Both frameworks guided our demo setup from MQTT authentication to Wazuh threat detection proving open-source tools meet international standards

Source: NIST CSF v1.1 (2018) | IEC 62443 (2018)

Case Study: JLR Industrial Breach (2025/2026)

- **Target: Jaguar Land Rover (**Global Production**).**
- **The Incident:** August 31, 2025 - A **"Digital Siege"** by the **Scattered Lapsus$** group.
- **The Breach:** Attackers exploited **Level 4 (Enterprise)** credentials to paralyze global production **Level 0-2 (Production)**.
- **Impact:** 3-weeks global shutdown; **£1.9 Billion (**$2.5B**)** total economic loss.
- **Solution:**
  - Wazuh: Would have detected the initial lateral movement at Level 4
  - Node-RED: Acts as a secure gateway to isolate the factory floor from IT breach
  - IEC 62443: Segmentation ensures that a "hacked office" does not mean a "stopped factory".
- **Reference: https://www.bbc.com/news/articles/c9wywvllq7wo**

Conclusion

1. Every connected device is an asset and a vulnerability.
2. Zero-cost tools protect factories at enterprise-grade level.
3. Authentication and encryption shield data end-to-end.
4. Real threats detected instantly live with Wazuh.
5. IEC 62443 and NIST CSF turn policy into working protection.

“A smart factory is only as secure as its weakest connected device - we built the tools to protect it.”

~Thank You